

Introduction

Baroid Core Homework 5

Assessment Feedback

Congratulations, you passed.

Total score: 72 out of 72, 100%

Question Feedback

Question

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Using the following diagram, complete the Waste Management Hierarchy

The diagram illustrates the Waste Management Hierarchy as a red funnel divided into six numbered levels. To the left of the funnel, the word **MINIMIZATION** is written vertically. To the right, a vertical arrow points upwards from **Most Expensive & Energy Consuming Options** to **Most Preferred Options**.

Level	Option	Score
1	Reuse	2
2	Disposal of Waste	6
3	Separation of waste	5
4		
5		
6		

Recycle

3 ▼

Recover

4 ▼

Reduce

1 ▼

6 out of 6

Question

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Increasing solids content in a drilling fluid can result in several negative issues.

From the following list, select the most critical effects of improper solids control

- ☒ Increased fluid loss due high ECDs
- ☒ Stuck pipe due differential sticking
- ☐ Increase of ROP and lowering of rig time
- ☒ Poor cement jobs due thick filtercake
- ☐ Increased lubrication
- ☒ Increase of rig time by lowering ROP
- ☒ Increased fluid waste volumes
- ☐ Good cement jobs due thinner filtercake

5 out of 5

Question

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Which statements below reflect the role of solids control equipment?

- ☐ Solids Control equipment typically provides discharged solids that are under the permitted ROC% limit to allow discharging them overboard
- ☒ Solids control operations address the first step of the waste management hierarchy by reducing the amount of generated waste
- ☐ Solids control equipment treats and segregates all waste bi-products from the discharged waste streams
- ☐ Solids Control operations should only be used with expensive fluid systems
- ☒ Solid control equipment is purposed to optimizes the drilling fluid performance and maximize drilling efficiencies

2 out of 2

Question

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What purpose would solids control equipment serve when encountering formations at depths that provide a narrow margin of error allowed between the pore pressure and fracture gradient

- ☐ Increase penetration rates (ROP's)
- ☐ Provide better cementing jobs

- ☐ Lower probability of differential sticking
- ☒ Provide Lower Equivalent Circulating Densities (ECD's)
- ☐ Reduced torque and drag

1 out of 1

Question

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What purpose would solids control equipment serve if an operator emphasized the importance to reach all section target depths (TD) on or ahead of schedule:

- ☒ Increase penetration rates (ROP's)
- ☐ Provide better cementing jobs
- ☐ Lower probability of differential sticking
- ☐ Provide Lower Equivalent Circulating Densities (ECD's)
- ☐ Reduced torque and drag

1 out of 1

Question

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What is considered the most effective and economically viable option to remove solids from the drilling fluid?

- ☐ Dump & dilute
- ☐ Chemical Treatment
- ☒ Mechanical separation
- ☐ Gravitational Settling
- ☐ Dilution

1 out of 1

Question

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What are the **two** most important reasons for adding weighting material and maintaining sufficient mud density?

- ☒ Provide wellbore stability
- ☐ Transport Cuttings to Surface
- ☐ Suspend and release cuttings
- ☐ Protect against corrosion
- ☐ Provide hydraulic power to downhole tools
- ☒ Control subsurface pressures

2 out of 2

Question

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Barite should consist of the following physical characteristics?

- ☒ Micron size of particles is predominantly <74 microns
- ☒ Non-abrasive
- ☐ Chemically reactive
- ☒ High specific gravity
- ☒ Chemically inert
- ☐ Low specific gravity
- ☐ Micron size of particles is predominantly >74 microns

4 out of 4

Question

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What are the two most important purposes of maintaining sufficient rheological properties?

- ☒ Suspend and release cuttings
- ☒ Transport cuttings to surface
- ☐ Lubricate the bit
- ☐ Provide wellbore stability
- ☐ Provide hydraulic power to downhole tools

2 out of 2

Question

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What are the four criteria which solid particles are typically classified?

- ☐ Solid Particle color
- ☒ Solid Particle Specific Gravity (SG)
- ☐ Solid Particle mineralogy
- ☒ Solid Particle Reactivity
- ☐ Solid Particle shape
- ☐ Solid Particle strength
- ☒ Solid Particle Source
- ☒ Solid Particle Size

4 out of 4

Question

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Solid particle degradation will result in smaller solid particle size causing an increase in the:

- ☐ Solid particle distribution
- ☐ Solid particles density
- ☐ Solid particles reactivity
- ☐ Solid particles abrasiveness
- ☒ Solid particles surface area

1 out of 1

Question

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What characteristic of some shale formations causes hydration of the clay particles resulting in solid particle swelling?

- ☐ Particle size
- ☐ Particle density
- ☐ Particle shape
- ☐ Particle surface area
- ☐ Particle degradation
- ☒ Particle reactivity

1 out of 1

Question

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The two solid characteristics that pose the most difficult challenges to the fluids rheological properties and overall fluid performance are:

- ☒ Solid Reactiveness
- ☐ Solid abrasiveness
- ☐ Solid density
- ☐ Solid specific gravity
- ☒ Solid size

2 out of 2

Question

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Which type of clay formation exhibits the highest potential for swelling?

- ☐ Kaolinites
- ☒ Smectites
- ☐ Chlorites
- ☐ Illites
- ☐ All of the above

1 out of 1

Question

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Why do clay particles that are “inert” still pose adverse challenges to the drilling fluid properties? (select all that apply)

- ☒ The clay particles are so fine (small) making removal difficult for conventional solids control equipment
- ☐ They consist of mostly abrasive solid particles causing further erosion of equipment.
- ☐ They consist of mostly large solid particles resulting in increased solid surface area and less free liquid available to the mud
- ☒ They consist of mostly colloidal solid particles causing increased solid surface areas and less free liquid providing undesired viscosity of mud

2 out of 2

Question

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Higher than normal sand content present in drilling muds can result in:

- ☐ Excessive hydration of clays
- ☐ Excessive dilution as sand is difficult to mechanically remove
- ☒ Excessive equipment repair and pump maintenance costs
- ☐ Increased amount of colloidal solid particles

1 out of 1

Question

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The use of dilution only to reduce solids % provides what implications?

- ☐ Does not lower the solids % in the mud
- ☒ Excessive mud and waste volumes
- ☒ Increasingly excessive mud maintenance costs
- ☐ Fluid incompatibility with drilled solids

2 out of 2

Question

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Use of gravitational settling only to lower solids content provides what operational implications?

- ☐ Operational footprint constraints
- ☐ Only the ability to settle out and remove large solid particles
- ☐ Operational time constraints
- ☒ All of the above

1 out of 1

Question

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What following aspects significantly affect the settling velocity of a drilled solid? *(Select all that apply)*

- ☒ Solid particles specific gravity
- ☐ Solid particles shape
- ☒ Solid particles micron size
- ☐ Solid particles color
- ☒ Fluid density
- ☒ Fluid Viscosity
- ☐ Fluid loss into the formation
- ☒ Amount of G-force applied

5 out of 5

Question

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Chemical additions of thinners / dispersants to repair and overcome undesired increased rheological properties is referred to as:

- ☐ Coagulation
- ☒ De-flocculation
- ☐ Inhibition
- ☐ Encapsulation
- ☐ Flocculation

1 out of 1

Question

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Encapsulation involves the additions of high molecular weight polymers to coat and preserve the drilled solids in order to prevent issues related to reactive solids

1 out of 1

Question

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A chemical treatment used to enhance the solids removal efficiencies, also known as *dewatering* typically consists of:

- a) The use of chemical additions to neutralize the negative charge of the solid particles resulting in solid particle aggregation is referred to as
- b) The use of long chain polymer additions to envelope and bind the neutralized solid aggregates together providing larger solid clusters is referred to as

2 out of 2

Question

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When drilling reactive formations prevention is generally accomplished by utilizing fluid systems that are formulated to incorporate the chemistry needed to prevent excessive dispersion & hydration of solid particles.

This is an example of:

- ☐ Emulsification
- ☒ Inhibition
- ☐ Coagulation
- ☐ De-flocculation
- ☐ Flocculation

1 out of 1

Question

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What are the categories that Baroid's BSS uses to classify their offered equipment and services?

- ☒ Filtration (wellbore cleanup) Equipment & Services
- ☒ Solids Control & Fluid Treatment Equipment & Services
- ☐ Waste Disposal & Transportation Equipment & Services
- ☒ Waste Handling & Transportation Equipment & Services
- ☒ Waste Treatment & Disposal Equipment & Services
- ☐ Waste Treatment & Transportation Equipment & Services
- ☐ Spacers & Fluid Mixing equipment
- ☐ Solids Solutions & Fluid Treatment Equipment & Services

4 out of 4

Question

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What section of the surface mud pit system are the solids control devices located? —

- ☒ Return section
- ☐ Suction section
- ☐ Additions section
- ☐ Testing section
- ☐ Reserves section

1 out of 1

Question

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A **Flow divider** is a device often installed on the flow line to ensure equal distribution of the circulated drilling fluid and transported solids over the shakers

1 out of 1

Question

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A **Gumbo removal device** is a preliminary screening mechanism utilized when drilling adolescent unconsolidated shale's causing excessive solid hydration that would obstruct the shaker performance

1 out of 1

Question

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What is the typical sequenced arrangement of conventional solids control equipment for a weighted system? —

- ☐ Shake shaker, desilter, desander, & centrifuge
- ☐ Shale shaker, centrifuge, desander
- ☒ Shale shaker, Mud Cleaner, Low Speed Centrifuge, High Speed Centrifuge
- ☐ Shale Shaker, desander, desilter & centrifuge
- ☐ Shale shaker, centrifuge, desander & desilter

1 out of 1

Question

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What is the typical sequenced arrangement of conventional solids control equipment for an un-weighted system?

- ☐ Shake shaker, desilter, desander, & centrifuge
- ☐ Shale shaker, centrifuge, desander
- ☐ Shale shaker, Mud Cleaner, Low Speed Centrifuge, High Speed Centrifuge
- ☒ Shale Shaker, desander, desilter & centrifuge
- ☐ Shale shaker, centrifuge, desander & desilter

1 out of 1

Question

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What is the angled settling pit located directly below the underbelly of the primary shakers?

- ☐ Trip tank
- ☐ Slugging pit
- ☒ Sand trap
- ☐ Return pit
- ☐ Coarse tank

1 out of 1

Question

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Which piece of Baroid Surface Solutions equipment is the first line of defense in solids control?

- ☒ Shale shaker
- ☐ Centrifuge
- ☐ Dryer
- ☐ Flow divider

1 out of 1

Question

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What conventional solids control equipment removes solids based on the principle of accelerated sedimentation?

- ☒ Hydrocyclones (desander and desilter)
- ☐ Shale shakers
- ☐ Cutting dryers
- ☒ Decanting centrifuge
- ☐ All of the above

2 out of 2

Question

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What separation method is used when using a Shale Shaker?

- ☐ Gravity
- ☒ Filtering
- ☐ Accelerated sedimentation
- ☐ Centrifugal force
- ☐ Desorption

1 out of 1

Question

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A “stand alone” Hydrocyclone is better known as:

- ☒ Desander
- ☐ Centrifuge
- ☐ Degasser
- ☒ Desilter
- ☐ All of the above

2 out of 2

Question

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A mud cleaner combines what **TWO** methods to remove solids from the drilling mud?

- ☒ Filtering
- ☐ Gravitational settling
- ☒ Accelerated Sedimentation
- ☐ Accelerated Degradation

2 out of 2

Question

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What piece of conventional solids control equipment is purposed to remove fine to ultra-fine solids less than 20 microns?

- ☐ Shale Shaker
- ☐ Desander

- ☐ Flow divider
- ☒ Centrifuge

1 out of 1

Question

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A mud cleaner typically is used in place of stand-alone hydrocyclones on what type of mud systems?

- ☐ Un-weighted systems
- ☐ Completions Brine
- ☒ Weighted systems
- ☐ Water based systems
- ☐ Non Aqueous fluid systems

1 out of 1

Question

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List the primary applications for centrifuges utilized by BSS

- ☒ Max solids removal
- ☐ Particle size reduction
- ☒ Rheology control (barite recovery)
- ☒ Chemically enhanced centrifugation (dewatering)
- ☐ Rheology increase
- ☐ None of the above

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